

HamidReza Saadi Dadmarzi

Education

Koç University, Istanbul, Turkey

2023/11 – now

M.Sc. in Computer Science

Research focus: Password-based authentication systems (MPC), Privacy-preserving ML and Post-Quantum Cryptography.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

2017/09 – 2020/08

M.Sc. in Applied Mathematics

Research focus: Quantum error correcting codes, Surface codes, information theoretic cryptography.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

2018/09 – 2020/08

Minor in Physics.

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

2014/09 – 2017/08

B.Sc. in Applied Mathematics

Ranked in the top 5% among all students.

Publications & Research Projects

Publications / Preprints:

- **UpSPA: Secure and Updatable Single Password Authentication** (submitted to ESORICS 2026), (ePrint version, code)
- **RoUpSPA: Robust Updatable Single Password Authentication** (preprint, to be submitted to ACM TOPS)

Selected Research Projects:

- **QuasarSSO: A Post-Quantum Password-Protected SSO System**
- **AEGIS: A Mechanizable Framework for Post-Quantum IoT Authenticated Key Exchange**
- **SpectraDP: Spectral Trade-offs for Communication-Efficient Differentially Private Learning**

Professional Experience

BTL Coin Crypto Exchange, Istanbul, Turkey /Remote

2021/05 – 2023/10

*Developed personalized market screeners and automated crypto asset management bots using **Golang** and **Docker**.*

Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

2020/10 – 2021/03

*Optimized quantum kernels for neural networks using **Qiskit** and **PyQuil**.*

Paid University Project, Tehran, Iran

2019/05 – 2020/07

*Developed a data warehouse and integrated data using **Excel** and **Tableau**.*

Honors & Awards

- Awarded double major recognition from the Exceptional Talent Center at the university.
- Admitted to the Master of Science program without a national entrance exam, recognized by the Exceptional Talent Center.
- Ranked in the top 5% among all students in the B.Sc. program.

Skills

- **Languages:** Persian (Native), English (Working Professional Proficiency), Turkish (Elementary).
- **Programming:** Rust, Python, Go (Golang), MATLAB.
- **Typesetting:** $\text{\LaTeX}$.
- **Specific libraries:** Qiskit, PyQuil, Lattigo.
- **Optimization:** `scipy.optimize`, CVXPY.

Academic Services

- **External reviewer for:** IEEE TIFS (2024–present), ACM ASIACCS (2024, 2026), ES-ORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025)
- Mentored two interns during Summer 2025; mentoring four interns from Fall 2025 through Summer 2026.

Teaching Experience

Amirkabir University of Technology

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|---|-----------------------|
| • Time Series Analysis | Fall 2019 |
| • Multivariable Calculus | Spring 2019 |
| • Introduction to C Programming | Fall 2018 |
| • Partial Differential Equations | Fall 2018 |
| • Foundation of Matrix and Linear Algebra | Spring 2018 |
| • One Variable Calculus | Fall 2017 |
| • Introduction to Probability | Spring 2017 |
| • General Physics (Electricity and Magnetism) | Spring 2017 |
| • Foundation of Mathematical Analysis | Spring 2017 |
| • Introduction to Set Theory and Logic | Fall 2016–Spring 2018 |

Koç University

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|--------------------------------------|--------------------------|
| • Introduction to Python Programming | Fall 2023 |
| • Discrete Structures | Spring 2024 |
| • Data Structures & Algorithms | Fall 2024 |
| • Computer Networks & Security | Spring 2025, Spring 2026 |
| • Modern Cryptography | Fall 2025 |

Conferences & Workshops

- 5th and 6th Frontiers in Mathematical Sciences Conferences, IPM, Tehran, July 2017 and July 2018.
- Workshop on Mathematics of Information-Theoretic Cryptography, IMS, Singapore, September 2016.

References

- Prof. Alptekin Küpçü ([link](#), [email](#))
- Dr. Erfan Salavati ([link](#), [email](#))
- Prof. Masoud Pourmahdian ([link](#), [email](#))